Mid 2019

Tags

1. a) How are collisions handled in Back2F (back to frequency) protocol? Explain with appropriate contention diagrams. 6
   b) Explain in brief a proposed channel access mechanism which is most similar to CSMA/CA and uses deterministic back-off. 7
   c) What is the difference between partial ordering and total ordering? 3.66

a)

Back to Frequency (B2F) is a channel access mechanism used in some wireless communication systems, such as RFID (radio frequency identification) networks, to avoid collisions when multiple devices attempt to transmit on the same frequency.

In B2F, devices that detect a collision on the communication channel are required to switch to a different frequency to re-transmit their data, using a pre-determined algorithm that determines the new frequency. The algorithm typically involves selecting a frequency that is different from the original one, based on a predefined pattern or formula. This helps ensure that the devices do not re-transmit on the same frequency and potentially cause another collision.

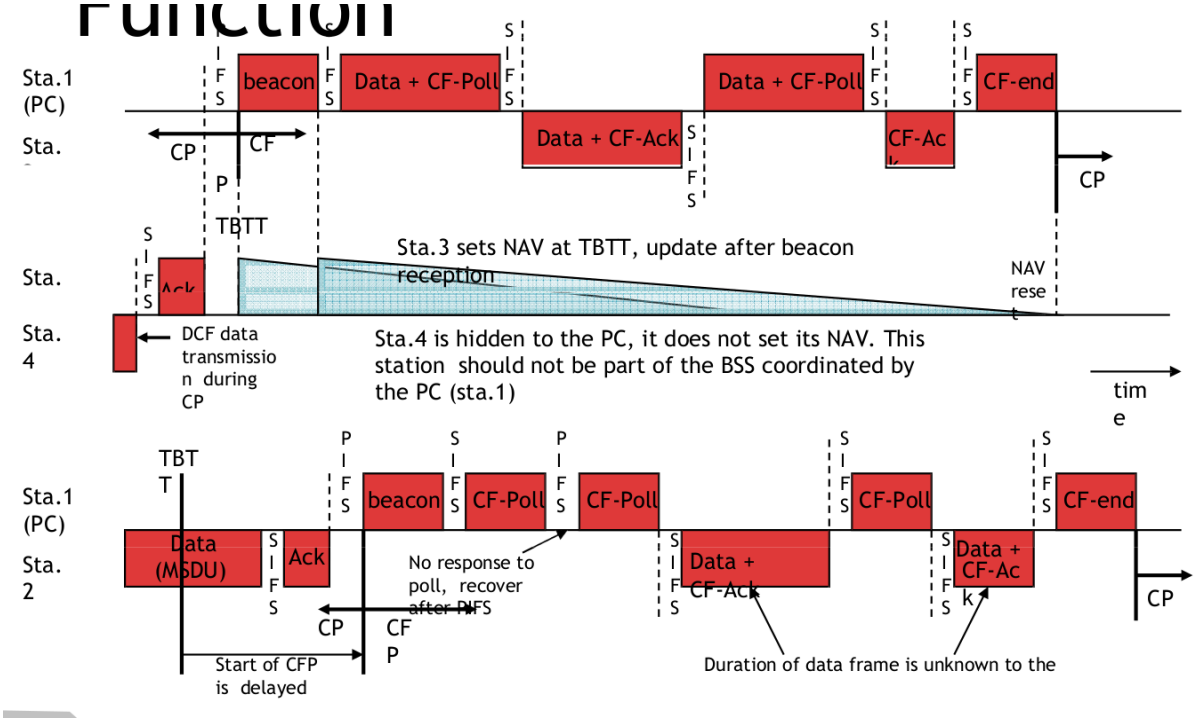
b) Deterministic backoff is a channel access mechanism used in communication networks to reduce the likelihood of collisions when multiple devices attempt to transmit data simultaneously. It involves a device waiting for a random amount of time before attempting to transmit again, with the length of the wait time increasing after each unsuccessful attempt, according to a predetermined algorithm or formula.

PCF is a similar channel access mechanism

QOS using PCF:

- The Point Coordination Function (PCF) let station have priority access to the radio channel, coordinated by a station called Point Coordinator (PC).
- PC typically resides in the AP.
- The PCF has higher priority than the DCF, because the period during which the PCF is used is protected from the DCF access by the NAV.

- The time during which (PCF) 802.11 stations operate is divided into repeated periods, called superframes. It includes CFP, CP.
- During the CFP, the PCF is used for accessing the channel, while the DCF is used during the CP



- Station 1 is the PC and polls station 2.
- Station 3 detects the beacon frame and updates the NAV for the whole CFP.
- CFP may be delayed (lower part of the figure).
- PC has data pending for station 2, combining data and poll frame into the data frame.
- The PC continues polling other stations until the CFP expires. No idle period longer than PIFS occurs during CFP.
- A CF-End control frame is transmitted to signal the end of the CFP.
- The time the beacon is delayed from TBTT determines the delay of the transmission of time- bounded MSDUs that have to be delivered in the CFP.

- The duration of the MSDU delivery after polling is not under the control of the PC, which reduces the QoS provided to other stations that are polled during the rest of the CFP.

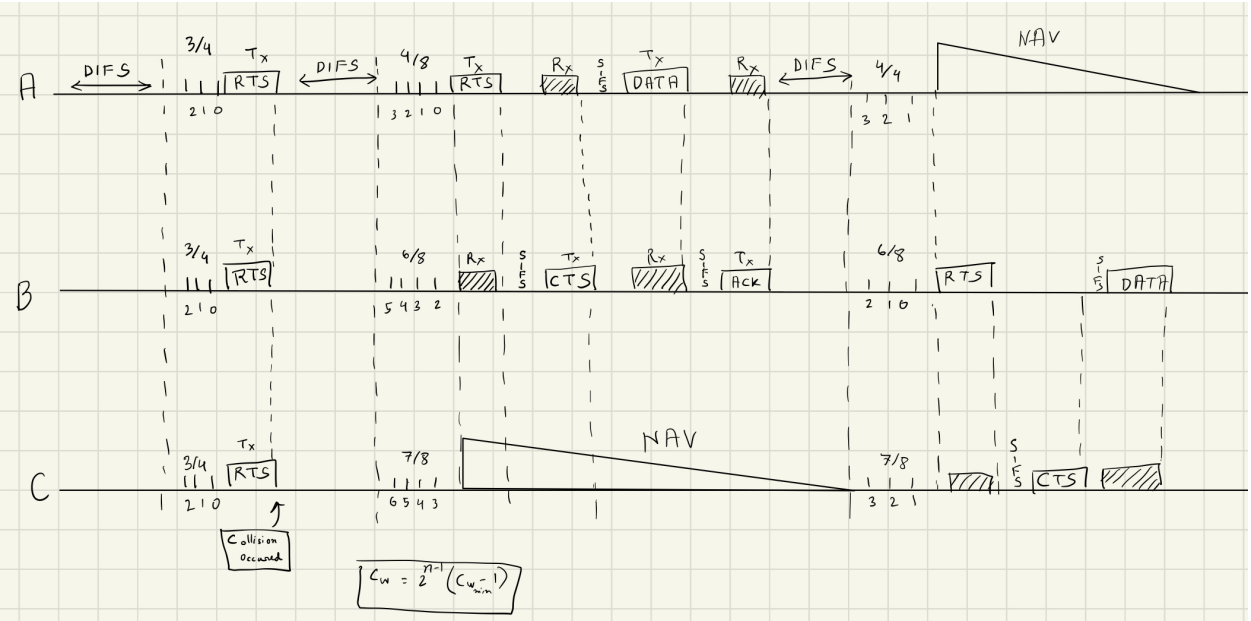
c)

In distributed computing and networking, ordering refers to the arrangement of events that occur across different nodes in a system

Partial ordering is a relationship that specifies only a partial ordering of events in a distributed system. In other words, it means that not all events can be ordered relative to one another. In a partially ordered system, events can be concurrent, meaning that they occur at the same time, and there is no way to determine which one happened first.

Total ordering, on the other hand, is a relationship that specifies a total ordering of events in a distributed system. This means that all events can be ordered relative to one another, and there is a clear sequence in which events occurred.

2. a) Illustrate a timeline diagram of a successful retransmission (a collision and then a successful transmission) in the current WLAN protocol (802.11) when 3 stations are trying to access the channel at the same time. Usage of RTS and CTS is optional. 6



b) From the given Figure 1, identify which of the nodes are "exposed" and which are "hidden". 3

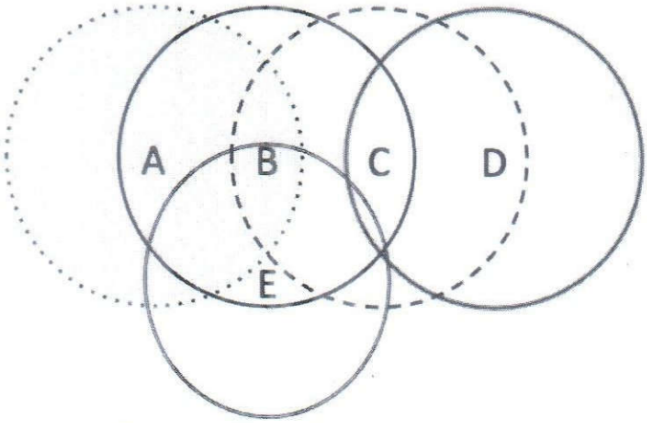
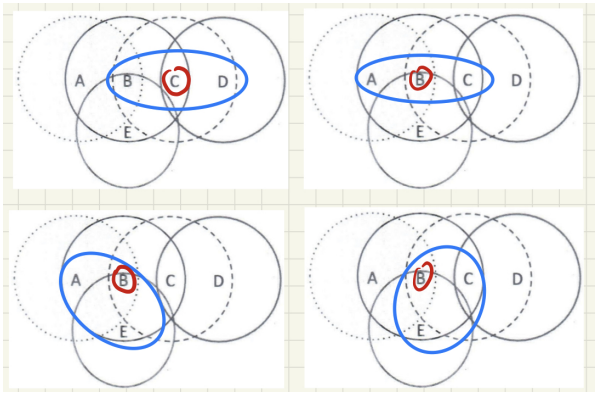


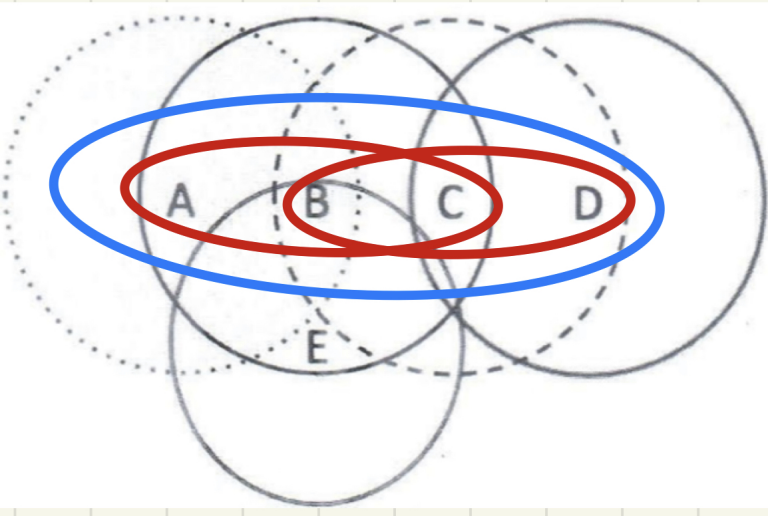
Figure 1: Figure for question 2 (b)



Hidden Stations

Table:

Sender 1	Sender 2	Reciever
B	D	C
A	C	B
E	C	B
A	E	B



Exposed Stations B & C

c) Does 802.11 (WLAN) solve the problems of Hidden and Exposed Terminal? Give appropriate reason. How does CSMA/CN contribute to solve these problems? 3 + 4.66

The hidden and exposed terminal problem is solved by the MAC (medium access control) layer protocol IEEE 802.11 RTS/CTS

Hidden Station:

To solve the hidden station problem, the IEEE 802.11 standard for wireless LANs incorporates a mechanism known as the Request to Send/Clear to Send (RTS/CTS) protocol. When a station wants to transmit data to another station, it first sends a small RTS frame to the access point or the other station, indicating that it wants to send data. The receiving station responds with a Clear to Send (CTS) frame, which indicates that it is ready to receive the data. Other stations that may be hidden from the transmitting station can also hear the CTS frame and will refrain from transmitting during the ensuing data transfer.

Exposed Station:

a station refrains from transmitting data even though it could do so without causing interference with ongoing transmissions. This occurs when a station senses that another station is transmitting on the shared medium, even though the other station is not actually interfering with the transmission of the first station.

To solve the exposed station problem, the IEEE 802.11 standard for wireless LANs incorporates a mechanism known as the Carrier Sense Multiple Access with Collision

Avoidance (CSMA/CA) protocol. In this protocol, a station first listens for ongoing transmissions on the shared medium before it starts transmitting. If it detects that the medium is busy, it defers its transmission until the medium becomes free.

However, the standard also includes a feature called the virtual carrier sensing mechanism, which allows a station to detect the presence of other stations even if they are not directly interfering with its transmission. When a station wants to transmit data, it sends out a small signal called the Network Allocation Vector (NAV) to other stations, indicating that it intends to use the shared medium for a certain amount of time. Other stations that hear the NAV signal know to refrain from transmitting for that period of time, even if they cannot sense any ongoing transmission from the first station.

By using the CSMA/CA protocol and the virtual carrier sensing mechanism, the IEEE 802.11 standard ensures that stations do not hold back on transmitting data even when there are other stations nearby that are transmitting, but not interfering with their transmissions. This helps to improve the efficiency and fairness of wireless communications.

3.	a)	What is HiBo? How does it minimize traditional contention time?	4+3
	b)	PC protocols have two phases – Learning phase and Transmission phase. What happens in each phase? Provide necessary figures.	9.66

a)

HiBo is a channel access mechanism designed to reduce contention time in wireless networks. It is a hybrid approach that combines the features of the Contention Window (CW) and the Busy Tone Avoidance (BTA) protocols, which are commonly used in wireless networks.

Traditional contention time in wireless networks can be high because of the randomness of the CW protocol, where stations pick a random backoff time before transmitting. This can lead to collisions and wasted transmission attempts, which results in an increase in contention time.

HiBo minimizes the traditional contention time by reducing the randomness of the backoff time used by the CW protocol. It does this by introducing a deterministic backoff time that is based on the round-trip time of the data packet. The HiBo algorithm also makes use of the BTA protocol to further reduce the contention time by avoiding collisions during the actual transmission of the data packet.

In the HiBo protocol, when a station wants to transmit a packet, it first sends out a request to the access point (AP). The AP then responds with a Grant packet that includes the deterministic backoff time. The station then waits for the specified time before transmitting the data packet. During the transmission, the BTA protocol is used to detect any ongoing transmissions on the shared medium and avoid collisions.

By using a deterministic backoff time and the BTA protocol, the HiBo mechanism minimizes the traditional contention time in wireless networks, leading to improved network efficiency and reduced delays.

b)

The PC (Persistent Carrier Sensing) protocol is a channel access mechanism used in wireless networks. It has two phases: the learning phase and the transmission phase.

1. Learning phase:
   During the learning phase, each station that wants to transmit data listens to the medium to determine if it is free for a fixed period of time known as the Learning Interval (LI). This allows the station to learn the state of the medium and adapt its behavior accordingly. If the medium is busy, the station increments a counter, called the Backoff Counter (BO), and waits for the medium to become free. The BO counter is decremented when the medium is found idle during the learning phase.
2. Transmission phase:
   During the transmission phase, a station that wants to transmit data sends a request-to-send (RTS) packet to the access point (AP). If the AP is available, it responds with a clear-to-send (CTS) packet. The station then waits for a predetermined time, known as the Short Interframe Space (SIFS), before transmitting the data packet. During the transmission, the station monitors the medium to ensure that no other station is transmitting data.

By using a learning phase and a transmission phase, the PC protocol improves the efficiency and reliability of wireless transmissions by minimizing collisions and wasted transmission attempts.

4. a) What is standardization? "Making any change to the current network protocol is a lengthy process without any certainty of approval" - Explain. 1.66 + 3

b) "The building blocks of wireless networks may or may not have centralized control. However, the presence of such a control is paramount to maintain Quality of Service (QoS)." – Express in terms of networking terminologies. 6

c) The backbone network of an ESS can be a variety of protocols. The distribution system (DS) connects several BSS together. Can the backbone network (or part of it) be compared to the Internet (a service view)? If so, how? 6

a)

Standardization refers to the process of developing and adopting technical standards that define the specifications, guidelines, and procedures for various technologies, products, and services. The goal of standardization is to ensure interoperability, compatibility, reliability, and quality of products and services across different vendors, platforms, and systems.

The process typically involves several stages, including the development of a draft proposal, review and feedback by technical experts and interested parties, revisions and refinements, and final approval. This process can take several years and involves multiple rounds of feedback and revision.

The approval of a new standard is not guaranteed and depends on various factors, including technical feasibility, market demand, and consensus among stakeholders. The standardization process requires a high degree of collaboration, coordination, and compromise among stakeholders to ensure that the resulting standard meets the needs of the industry, users, and other stakeholders.

b)

The presence of centralized control is paramount to maintain Quality of Service (QoS) in wireless networks. QoS refers to the ability of a network to provide reliable and predictable performance to different types of traffic, such as voice, video, and data. In a wireless network, QoS is particularly important due to the inherent variability and unpredictability of the wireless medium.

By having a central entity that can monitor and manage network traffic, allocate network resources, and make real-time decisions to optimize network performance, wireless networks can ensure that QoS requirements are met. Without centralized control, wireless networks may be subject to interference, congestion, and other factors that can degrade network performance and impact QoS.

c)

The distribution system (DS) connects multiple Basic Service Sets (BSS) in the wireless network, allowing users to move between different BSSs without losing their connection. The DS can also include components such as bridges or routers that allow communication between different subnets.

In terms of a service view, the backbone network of an ESS could be compared to the Internet in the sense that both provide a high-speed, reliable network infrastructure for transmitting data between different endpoints. However, the Internet is a much larger and more complex network that connects a vast array of different networks, devices, and services, while the backbone network of an ESS is typically a more limited, closed network designed specifically for the wireless network's needs.

Additionally, while the Internet is a global, public network that provides access to a wide range of services and resources, the backbone network of an ESS is typically a private, closed network that is only accessible to authorized users and devices within the wireless network.

